

(potentially fatal), should lead to immediate discontinuation of administration.

Adenosine may trigger convulsions in patients who are susceptible to convulsions. In patients with history of convulsions/seizures, the administration of adenosine should be carefully monitored.

Because of the possible risk of torsades de pointes, ADENORYTHM SOLUTION FOR INJECTION should be used with caution in patients with a prolonged QT interval, whether this is drug induced or of metabolic origin. ADENORYTHM SOLUTION FOR INJECTION is contraindicated in patients with Long QT syndrome (see section Contraindications).

Adenosine may precipitate or aggravate bronchospasm (see sections Contraindications and Adverse Reactions).

ADENORYTHM SOLUTION FOR INJECTION contains 9mg of sodium chloride per 1ml of solution (corresponding to 3.54mg (0.15mmol) of sodium per 1ml of solution). To be taken into consideration by patients on a controlled sodium diet.

ADENORYTHM SOLUTION FOR INJECTION may trigger atrial arrhythmias and thus might lead to ventricular acceleration in children with Wolff-Parkinson-White (WPW) syndrome.

The efficacy of intraosseous administration in children has not been established.

Use in pregnancy & lactation:

Pregnancy:

There are no or limited amount of data from the use of adenosine in pregnant women. Animal studies are insufficient with respect to reproductive toxicity. Adenosine is not recommended during pregnancy unless the physician considers the benefits to outweigh the potential risks.

Lactation:

It is unknown whether adenosine metabolites are excreted in human milk. ADENORYTHM SOLUTION FOR INJECTION should not be used during breast-feeding.

Drug Interactions:

Dipyridamole inhibits adenosine cellular uptake and metabolism, and potentiates the action of adenosine. In one study, dipyridamole was shown to produce a 4-fold increase in adenosine actions. Asystole has been reported following concomitant administration.

It is therefore suggested that ADENORYTHM SOLUTION FOR INJECTION should not be administered to patients receiving dipyridamole; if use of ADENORYTHM SOLUTION FOR INJECTION is essential, dipyridamole should be stopped 24 hours beforehand, or the dose of ADENORYTHM SOLUTION FOR INJECTION should be greatly reduced (See section Special Warnings & Precautions).

Aminophylline, theophylline and other xanthines are competitive adenosine antagonists and should be avoided for 24 hours prior to use of adenosine.

Food and drinks containing xanthines (tea, coffee, chocolate and cola) should be avoided for at least 12 hours prior to use of adenosine.

ADENORYTHM SOLUTION FOR INJECTION may interact with drugs tending to impair cardiac conduction.

Incompatibilities:

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

Adverse Reactions:

These side effects are generally mild, of short duration (usually less than 1 minute) and well tolerated by the patient. However severe reactions can occur.

Methylxanthines, such as IV aminophylline or theophylline have been used to terminate persistent side effects (50-125mg by slow intravenous injection).

Adverse events are ranked under the heading of the frequency: Very common (>1/10), Common (>1/100, <1/10), Uncommon (>1/1,000, <1/100), Rare (>1/10,000, <1/1,000), Very rare (<1/10,000), Not known (cannot be estimated from available data).

Cardiac Disorders:

Very common:

Bradycardia; sinus pause, skipped beats; atrial extrasystoles; atrio-ventricular block; ventricular excitability disorders such as ventricular extrasystoles, non-sustained ventricular tachycardia.

Uncommon:

Sinus tachycardia; palpitations.

Very rare:

Atrial fibrillation; severe bradycardia not corrected by atropine and possibly requiring temporary pacing; ventricular excitability disorders including ventricular fibrillation and torsade de pointes (see section Special Warnings and Precautions).

Not known:

Hypotension sometimes severe; asystole/cardiac arrest, sometimes fatal especially in patients with underlying ischemic heart disease/ cardiac disorder (see section Special Warnings and Precautions).

Nervous System Disorders:

Common:

Headache; dizziness, light-headedness.

Uncommon:

Head pressure.

Very rare:

Transient and spontaneously rapidly reversible worsening of intracranial hypertension.

Not known:

Loss of consciousness/syncope; convulsions, especially in predisposed patients (see section Special Warnings and Precautions).

Eye Disorders:

Uncommon:

Blurred vision.

Respiratory, Thoracic and Mediastinal Disorders:

Very common:

Dyspnea (or the urge to take a deep breath).

Uncommon:

Hyperventilation.

Very rare:

Bronchospasm (see section Special Warnings and Precautions).

Not known:

Respiratory failure (see section Special Warnings and Precautions); apnea/respiratory arrest.

Cases of respiratory failure, bronchospasm, apnea, and respiratory arrest with fatal outcome have been reported.

Gastrointestinal Disorders:

Common:

Nausea.

Uncommon:

Metallic taste.

Not known:

Vomiting.

Vascular Disorders:

Very common:

Flushing.

Immune System Disorders:

Not known:

Anaphylactic reaction (including angioedema and skin reactions such as urticaria and rash).

General Disorders and Administration Site Conditions:

Very common:

Chest pressure/pain, feeling of thoracic constriction/oppression.

Common:

Burning sensation.

Uncommon:

Sweating; feeling of general discomfort/weakness/pain.

Very rare:

Injection site reactions.

Psychiatric Disorders:

Common:

Apprehension.

Overdosage:

Overdosage would cause severe hypotension, bradycardia or asystole. The half life of adenosine in blood is very short, and side effects (when they occur) would quickly resolve. Administration of IV aminophylline or theophylline may be needed. Pharmacokinetic evaluation indicates that methylxanthines are competitive antagonists to adenosine, and that therapeutic concentrations of theophylline block its exogenous effects.

Storage Conditions:

Store below 30°C. Protect from light. Do not refrigerate as crystallisation may occur. The solution should be clear at time of use. Keep out of reach of children. Jauhi daripada kanak-kanak. The product is for single use only. Any portion of the vial, not used at once after opening, should be discarded. The product should be inspected visually for particulate matter and colouration prior to administration. Where the visual appearance of the product may have changed, the vial should be discarded. Any unused product or waste material should be disposed of in accordance with local requirements.

Dosage Form and Packaging Available:

ADENORYTHM SOLUTION FOR INJECTION is a sterile, clear, colourless, free from visible particulates solution for injection. It is supplied in a clear, transparent, colourless, Ph. Eur. Type I, 3mL capacity glass vial containing 2mL of solution. Each vial is closed with a chlorobutyl rubber closure and secured with a polypropylene/aluminium flip-off cap. A set of three vials is packed in a PVC holding tray. Two trays are packed into a cardboard carton with a product insert.

Packaging available: 6 vials X 2mL per box.

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Manufacturer: **VIANEX S.A. – PLANT A**, 12th km Athens-Lamia National Road, 14451 Metamorphossi, Attiki, Greece.